

JKBOSE 2026

CLASS 11TH

Discovering Tut

LECTURE-3

ENGLISH

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Topics To Covered

- 1 ...lec.3. (discovering Tut: The Saga Continuous)
- 2 ... Summary and Textual Questions
- 3 ...
- 4 ...

Discovering Tut: the Saga Continues *A.R. Williams*

Today diagnostic imaging can be done with computed tomography, or CT, by which hundreds of X-rays in cross section are put together like slices of bread to create a three-dimensional virtual body. What more would a CT scan reveal of Tut than the X-ray? And could it answer two of the biggest questions still lingering about him how did he die, and how old was he at the time of his death?

King Tut's ^{death} demise was a big event, even by royal standards. He was the last of his family's line, and his funeral was the death rattle of a dynasty. But the particulars of his passing away and its aftermath are unclear.



Death rattle: the indication of the end of the dynasty

Discovering Tut: the Saga Continues *A.R. Williams*



Amenhotep III — Tut's father or grandfather — was a powerful pharaoh who ruled for almost 40 years at the height of the eighteenth dynasty's golden age. His son Amenhotep IV succeeded him and started one of the strangest periods in the history of ancient Egypt. The new pharaoh promoted the worship of the Aten the sun disk, changed his name to Akhenaten, or 'servant of the Aten,' and moved the religious capital from the old city of Thebes to the new city of Akhetaten, known now as Amarna. He further shocked the country by attacking Amun, a major god, destroying his images and closing his temples. "It must have been a horrific time," said Ray Johnson, director of the University of Chicago's research centre in Luxor, the site of ancient Thebes. "The family that had ruled for centuries was coming to an end, and then Akhenaten went a little wacky."



Wacky: behaving in an odd way

Discovering Tut: the Saga Continues *A.R. Williams*



After Akhenaten's death, a ^{unknown} mysterious ruler named Smenkhkare appeared briefly and exited with hardly a trace. And then a very young Tutankhaten took the throne - King Tut as he's widely known today. The boy king soon changed his name to Tutankhamun, 'living image of Amun,' and oversaw a restoration of the old ways. He ^{ruled} reigned for about nine years -and then died unexpectedly.



Discovering Tut: the Saga Continues *A.R. Williams*

Regardless of his fame and the ^{guess} speculations about his fate, Tut is one mummy among many in Egypt. How many? No one knows. The Egyptian Mummy Project, which began an ^{list} inventory in late 2003, has recorded almost 600 so far and is still counting. The next phase: scanning the mummies with a ^{movable} portable CT machine donated by the National Geographic Society and Siemens, its manufacturer. King Tut is one of the first, mummies to be scanned – in death, as in life, moving ^{like a king} regally ahead of his countrymen.



Discovering Tut: the Saga Continues *A.R. Williams*

A CT machine scanned the mummy head to toe, creating 1,700 digital X-ray images in cross section. Tut's head, scanned in 0.62 millimetre slices to register its intricate structures, takes on eerie detail in the resulting image. With Tut's entire body similarly recorded, a team of specialists in radiology, forensics, and anatomy began to investigate the secrets that the winged goddesses of a gilded burial shrine protected for so long.



Discovering Tut: the Saga Continues *A.R. Williams*

The night of the scan, workmen carried Tut from the tomb in his box. Like pallbearers they climbed a ramp and a flight of stairs into the swirling sand outside, then rose on a hydraulic lift into the trailer that held the scanner. Twenty minutes later two men emerged, sprinted ^{ran very fast} for an office nearby, and returned with a pair of white plastic fans. The million-dollar scanner had quit because of sand in a cooler fan. "Curse of the pharaoh," joked a guard nervously.



Pallbearers: people who carry a coffin

Swirling: to spin or twist

Hydraulic lift: a lift that uses a machine to lift or move heavy objects with a pressure

Sprinted: ran at a high speed

Discovering Tut: the Saga Continues *A.R. Williams*

Eventually the substitute fans worked well enough to finish the procedure. After checking that no data had been lost, the technicians turned Tut over to the workmen, who carried him back to his tomb. Less than three hours after he was removed from his coffin, the pharaoh again rested in peace where the funerary priests had laid him so long ago. → 3300



Discovering Tut: the Saga Continues *A.R. Williams*



Back in the trailer a technician pulled up astonishing images of Tut on a computer screen. A grey head took shape from a scattering of pixels, and the technician spun and tilted it in every direction. Neck vertebrae appeared as clearly as in an anatomy class. Other images revealed a hand, several views of the rib cage, and a transection of the skull. But for now the pressure was off. Sitting back in his chair, Zahi Hawass smiled, visibly relieved that nothing had gone seriously wrong. "I didn't sleep last night, not for a second," he said. "I was so worried. But now I think I will go and sleep."



Pixels: a pixel is a single point in a graphic image

Spun: to turn around

Vertebrae: series of small bones which form a backbone

Discovering Tut: the Saga Continues *A.R. Williams*

By the time we left the trailer, descending metal stairs to the sandy ground, the wind had stopped. The winter air lay cold and still, like death itself, in this valley of the departed. Just above the entrance to Tut's tomb stood Orion – the ^{group of stars} constellation that the ancient Egyptians knew as the soul of Osiris, the god of the afterlife – watching over the boy king.

→ Osiris → God of the afterlife



QUESTION



What does Tutankhamun's tomb give us an insight into?

- 1 Egyptian wars
- 2 Tut's family tree
- 3 Ancient Egyptian beliefs and burial practices
- 4 Roman civilization

QUESTION



What was the atmosphere like when the CT scan was performed?

- 1 Stormy and windy
- 2 Calm and bright
- 3 Dark and silent with a cloudy sky ✓
- 4 Festive and colorful

QUESTION



What mystery surrounds Tutankhamun's death?

- 1 He died in battle
- 2 He was killed by a lion
- 3 His death is unknown and possibly suspicious
- 4 He died of old age

QUESTION



What nickname is given to Tutankhamun in modern media?

- 1 King of Gold
- 2 The Living Mummy
- 3 The Boy King ✓
- 4 ^{Last} The Lost Pharaoh

QUESTION



What do the funerary treasures found in Tut's tomb suggest?

- 1 He was poor
- 2 Egyptians believed in an afterlife
- 3 Tut was a slave
- 4 They wanted to bribe the gods

Tut's death and mystery

Tutankhamun, the last heir of a powerful royal family of Egypt's New Kingdom, died when he was still a teenager. His death was sudden and has always been a mystery. He was buried with unimaginable ^{valuable} riches – gold and treasures – which showed how much importance Egyptians gave to the afterlife. His tomb was discovered in 1922 by Howard Carter, and ever since, the young king's life and death have fascinated people. Many suspected he might have been murdered. Now, after more than 80 ²⁰⁰⁵ years, scientists prepared to investigate him with the help of modern technology – a CT scanner – that could provide a complete picture of his body and maybe recreate what he looked like.

Discovering Tut



The scene at Tut's tomb in 2005

It was 6 p.m. on 5 January 2005, at the Valley of the Kings. The evening was stormy – wind was strong, dust was rising, and dark clouds spread across the desert. Inside the tomb, Tut's body was taken out from the resting place and moved towards the CT machine. This machine was like a giant camera that could take hundreds of X-ray slices of his body. Earlier that day, many tourists had come to see the tomb, admired the paintings on the walls, and gazed at Tut's face on his golden coffin. The author points out the contrast – in the day the tomb was a tourist spot, and in the evening it turned into a scientific laboratory. Tut was about to be examined in a way no one from his time could have imagined.

Discovering Tut



Carter's discovery and what happened to Tut's body

The head of the Egyptian Supreme Council of Antiquities, Dr. Zahi Hawass, said Tut's mummy was in "bad condition because of what Carter did in the 1920s." Howard Carter, the British archaeologist, found Tut's tomb in 1922. Though the tomb had been robbed long ago, many objects were still there – golden collars, necklaces, amulets, rings, daggers, cups, and the most famous gold mask. These were meant to protect the king in the afterlife. There were also ordinary things – food, wine, games, clothes – to make him comfortable after death. But Carter faced a problem: the body had been cemented to the bottom of the coffin with hardened resins. Gentle methods failed. Even heating the coffin in the sun for hours did not work. Finally, Carter's men cut the mummy into pieces. They separated the head and all major joints. They then reassembled the body and placed it into a wooden box. Carter did it because he feared thieves might damage the body if he left it. But this rough treatment left the mummy badly damaged forever.

Modern archaeology and CT scanning

The author explains that archaeology has changed since Carter's time. Today archaeologists are not interested only in treasure but in knowledge – even small remains like seeds, hair, or pottery can give clues about history. In 1968, a medical X-ray showed that Tut's chest bones and sternum (breastbone) were missing. That raised new questions. Now with CT (computed tomography), a better technique was available. A CT scan takes many slices of X-ray images and combines them into a 3D picture, just like slices of bread make a loaf. With CT, scientists hoped to find out how Tut died, whether he was murdered or died naturally, and also his exact age at death.

Tut's dynasty and the Amarna period

The story now shifts to Tut's place in history. Tutankhamun was a descendant of a royal family that had ruled Egypt for centuries. His grandfather Amenhotep III ruled for almost 40 years. After him came Amenhotep IV, who shocked people by changing his name to Akhenaten. He promoted the worship of Aten, the sun god, and closed temples of the old god Amun. He even destroyed their images. He moved the capital from Thebes to a new city he built, Amarna. This religious revolution disturbed the people. Akhenaten's family was mysterious and troubled. After his death, a short-lived ruler Smenkhkare appeared, and then Tut, a child, came to the throne. Tut changed his name from Tutankhaten to Tutankhamun, showing he returned to the old religion of Amun. He ruled for nine years before dying suddenly. His death ended the dynasty and left behind many mysteries.

The Egyptian Mummy Project and the scan

In 2003, the Egyptian Mummy Project was started to study and record details of all ancient mummies. Tut was one of the first to be examined. A portable CT machine, donated by Siemens and the National Geographic Society, was brought to the Valley of the Kings. On 5 January 2005, Tut's body was scanned. It produced around 1700 images. The head was scanned in 0.62 mm slices, giving very clear details. But there was a small problem: sand had entered the cooler fan of the machine, and it stopped working for some time. People became nervous, and a guard jokingly said it was the "curse of the pharaoh." However, the team fixed the problem and finished the scan successfully. In less than three hours, Tut's body was put back in his tomb. The images showed his bones clearly, and scientists could rotate the head in any direction on a computer screen. Hawass looked relieved and proud that modern science had given new life to the study of the boy king.

Conclusion

The chapter ends with a sense of respect. Tut, the teenage king, died more than 3,300 years ago, but modern science is still chasing answers about his life and death. His story links the past and the present. Archaeology has moved from treasure hunting to careful study, and CT scanning may finally solve mysteries that even time could not erase.

QUESTION



King Tut's body has been subjected to repeated scrutiny.

ANSWER:

King Tut's body has been repeatedly examined because he died at a very young age under mysterious circumstances. Scholars and scientists have been curious to find the cause of his untimely death and gather more information about his life and reign.

QUESTION



Howard Carter's investigation was resented.

ANSWER:

Howard Carter's investigation was resented because he damaged the mummy during his examination. In his attempt to remove Tut's body from the coffin, he cut it into several pieces. This act was considered unscientific and disrespectful by many historians and scholars.

QUESTION



Carter had to chisel away the solidified resins to raise the king's remains.

ANSWER:

Carter had to chisel away the hardened resins because the body had become firmly glued to the coffin due to the application of large amounts of resin during burial. The sun's heat failed to loosen it, so Carter used a chisel and hammer to remove the remains.

QUESTION



Tut's body was buried along with gilded treasures.

ANSWER:

Tut's body was buried with gilded treasures because the ancient Egyptians believed in life after death. They thought that the king would need these items in the afterlife for comfort, protection, and royal status.

QUESTION



Here are some commonly used medical terms. Find out their meanings.

- CT scan
- MRI
- Tomography
- Autopsy
- Dialysis
- ECG
- Post mortem
- Angiography
- Biopsy

Handwritten signature

QUESTION



- **CT scan:** Short for Computerised Tomography, it is a specialised x-ray test that uses a series of X-ray images and a computer to create cross-sectional images of the body.
- **MRI:** Magnetic Resonance Imaging (MRI) is a diagnostic technique that employs magnetic fields and radio waves to generate a detailed image of the body's soft tissue and bones.
- **Tomography:** It refers to sectional imaging or sectioning using any type of penetrating wave, It is used in radiology, archaeology, and other scientific fields.

QUESTION



- **Autopsy:** It is a surgical procedure that entails dissecting a corpse to determine the cause and manner of death or to evaluate an injury for research purposes.
- **Dialysis:** It is the process of removing excess water and toxins from the blood in people whose kidneys have lost the ability to do so naturally.
- **ECG:** The electrocardiogram, also known as an ECG, is a diagnostic tool used to evaluate the electrical and muscular functions of the heart.

QUESTION



- **Post Mortem:** A corpse is examined or analysed soon after death to determine the cause of death.
- **Angiography:** It is a type of imaging tool used to see inside blood vessels and organs of the body, specifically arteries, veins, and heart chambers.
- **Biopsy:** It is a medical procedure in which surgeons, radiologists, or cardiologists extract cells or tissues for examination in order to determine the presence and extent of a disease

QUESTION



Howard Carter's investigation was resented.

ANSWER:

The statement means that King Tut, who was an important figure in life, continued to attract attention even in death. He was among the first mummies to undergo a modern CT scan, symbolizing how he remained significant in both life and after death through scientific and historical study.

QUESTION



The boy king changed his name from Tutankhaten to Tutankhamun.

ANSWER:

Tutankhaten changed his name to Tutankhamun to show his devotion to the god Amun and to distance himself from his father Akhenaten's unpopular monotheistic worship of Aten. This marked a return to traditional religious practices.

QUESTION



List the deeds that led Ray Johnson to describe Akhenaten as "wacky."

ANSWER:

Ray Johnson called Akhenaten "wacky" because of his radical reforms. Akhenaten rejected traditional Egyptian gods and introduced the worship of Aten, the sun disk. He moved the religious capital from Thebes to a new city, changed the art style, and even altered his name. These drastic changes disturbed Egyptian society and priesthood.

QUESTION



What were the results of the CT scan?

ANSWER:

The CT scan of King Tut's body provided detailed internal images. It showed that his head and bones were in good condition. No evidence of a fatal blow to the head or major bone injuries was found. Although the exact cause of death remained unclear, the scan helped rule out some theories and allowed further scientific investigation.

Thank
You